SMRTTECH 4AI3 - Artifical Intelligence and Machine Learning
Lab 6 - K-means & clustering

Instructor: Siqi Zhao

Due: Midnight 11:59PM, day before your next lab session

Please read and follow these instructions:

- Submit your lab on Avenue to Learn (A2L) before the due date (midnight of the day before your next lab session). The submission will be closed after the deadline.
- Do your lab exercises on your own, i.e., do not copy. If two lab reports (or some parts of them) are found to be copies of each other, then BOTH shall receive 0 mark.
- You may use Google Colab or any other Python programming platform of your own choice to run your programs.
- Please read the questions carefully and use AI / machine learning libraries only when specially required as part of the question. Also add comments in your code where ever necessary.
- Submit your working python source files (or Jupyter notebooks), **don't submit Zip file**, please upload your colab notebook link in the labs' submission dropbox comment section if you used Colab. Make sure to change the access setting to **"Anyone with the link."**

1 Exercises (20 Points)

Scikit-Learn provides a make_blobs () method to generate a clustering dataset. It is often used to visualize clustering and classification algorithms.

1. **Import make_blobs library to generate clustering dataset**

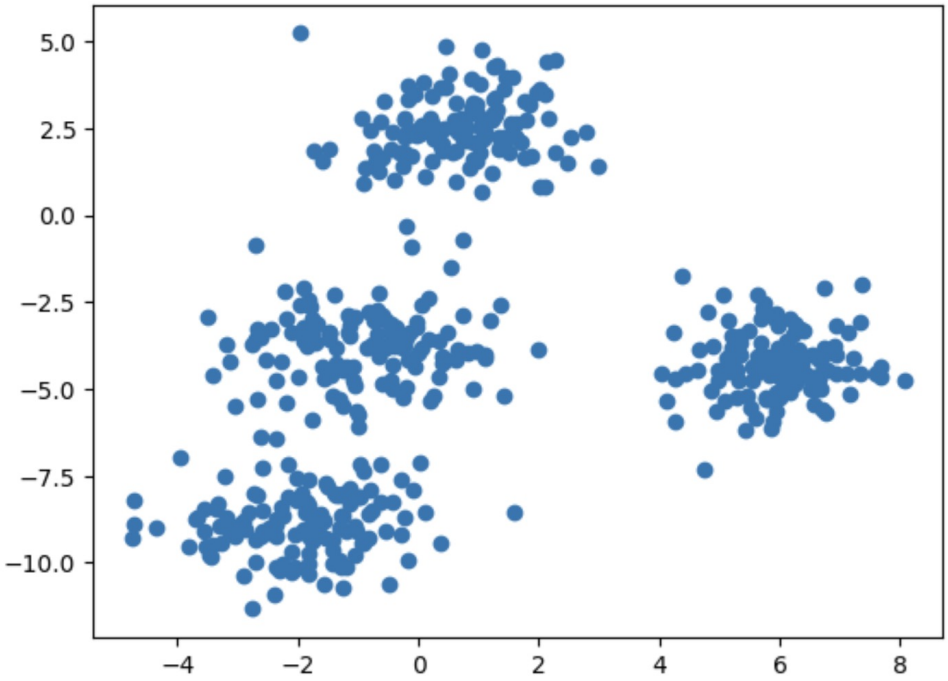
   from sklearn.datasets import make_blobs

2. **Assign X and y to the data points and the corresponding labels. The parameter n_samples determines how many points to be generated. The parameter centres determines how many clusters to be generated. And n_features controls how many features/attributes in the X. Please remain shuffle=True to generated shuffled dataset.**

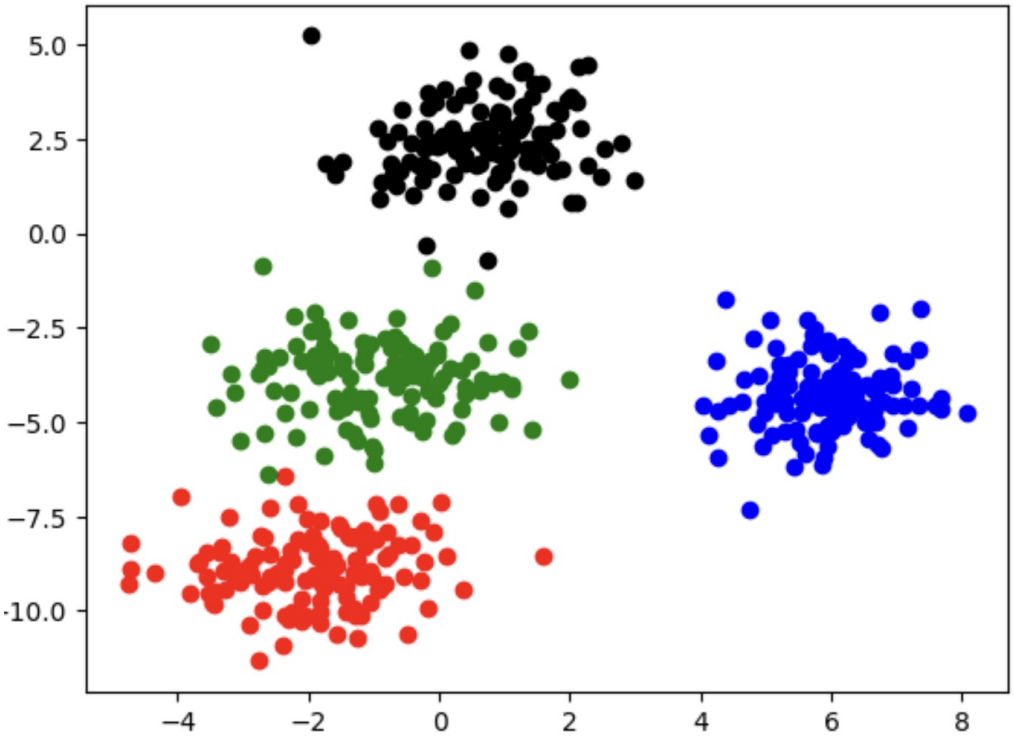
   X, y = make_blobs(centers=4, n_samples=500, n_features=2, shuffle=True, random_state=40)

3. **Take a look at the shape of X and y, we have 500 data points and 2 features.**

4. **Plot a scatter plot for the dataset we generate by using make_blobs()**

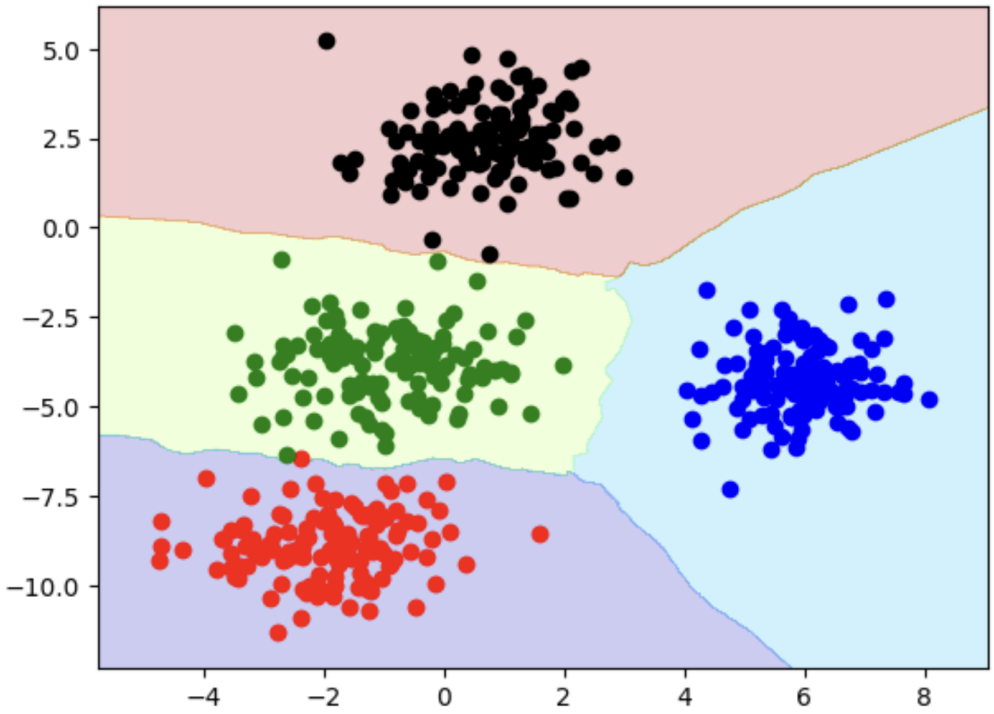


5. **Please generate a scatter plot again for this 2-D data. Use different colors to differentiate data points for different classes.**



6. **Split the data points into a training set and testing set using the train_test_split() method. Limit the size of the testing set to 20% with a fixed random state number.**

7. **Implement K-means algorithm from scratch without using sklearn library. And visualize classification boundary**



8. **Implement K-means algorithm with sklearn library. And visualize classification boundary**

9. **Implement KNN classifier by using sklearn.neighbors.KNeighborsClassifier(), fit the model with the training set and training labels, predict the result on test set and evaluate the model's accuracy. And visualize classification boundary**

10. **Briefly describe the difference between K-means and KNN algorithm**